

Biology and its relationship with other Sciences.

(ඒම විද්‍යාව සහ අනෙකුත් විද්‍යාවන් අතර ඇති සම්බන්ධතාව.)

- * To study Biology we should know other sciences like physics and chemistry.
- * cell membrane - structure of lipids and proteins - chemistry.
- * Release of energy during respiration of different chain.
- * Effect of light ~~on~~ on flowering - Absorption of different wave length of light.

Scientific method in Biology - ඒම විද්‍යාවේ විද්‍යාත්මක ක්‍රමය / මූලධර්මය.

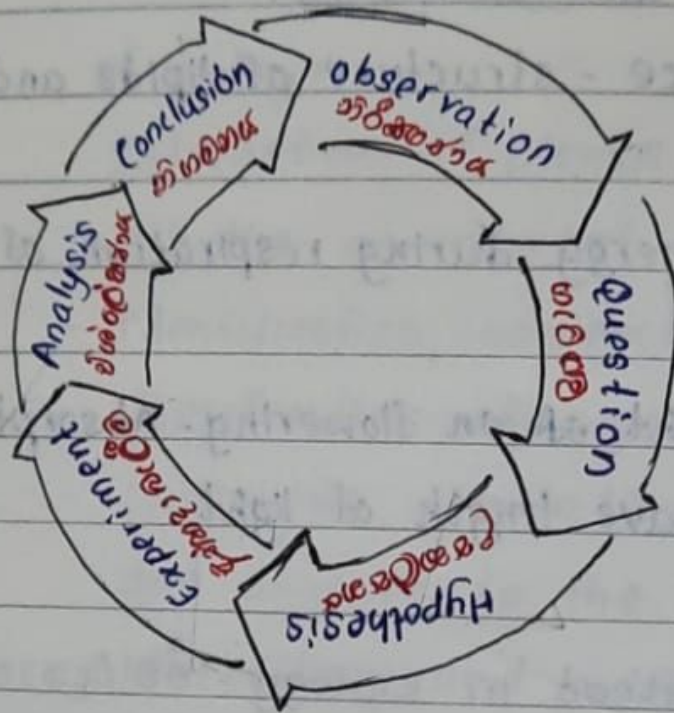
- * It is a difficult to find and answer without logical solution.

එය උනාමත් තර්කානුකූලව දෙයක් සොයාගැනීමට සහ පිළිතුරක් සොයාගැනීමට අපහසුය.

- * We need to define the problem, experiment it, analyze it and come to conclusion. අප ගැටළුව නිර්වචනය කර සහ අත්හදා බලා විශ්ලේෂණය කර නිගමනයට පැමිණිය යුතුය.

* It is done through scientific method which draws a definite conclusion.

ഈ കണ്ടുപിടുത്തം ശാസ്ത്രീയ രീതിയിൽ നിർവ്വഹിക്കപ്പെട്ടതാണ്. അതിനാൽ ഇതിൽ നിർവ്വചനാത്മകമായ ഒരു നിഗമനം ഉണ്ടാകുന്നു.



01) Observation : നിരീക്ഷണം.

* Most biological investigations - start with observation on structure, process, behavioral pattern.

അതായത് ജീവജാലങ്ങളുടെ ഘടന, പ്രക്രിയ, പെരുമാറ്റം, പ്രതികരണം, പ്രജനനം, പ്രവൃത്തി എന്നിവയെക്കുറിച്ച് നിരീക്ഷിക്കുന്നു.

ii) Defining a problem : അമിതമായി നിർമ്മാണ കിരീടം.

* The observations lead to questions.

നിർമ്മാണ പ്രക്രിയയെ എങ്ങനെ നോക്കി.

* What is this, how does it work, why does it work the way it does?

എന്തെല്ലാം കാര്യങ്ങൾ? അവയുടെ കാര്യങ്ങൾ എന്തെല്ലാം? അവയുടെ കാര്യങ്ങൾ എന്തെല്ലാം?

* This may necessitate further observations to be made.

ഒരു ജനതയെ ഉപയോഗിച്ച് നിർമ്മാണ പ്രക്രിയയെ നോക്കി.

iii) Hypothesis: പ്രയോജനപ്പെടുത്തൽ.

* On the basis of the facts known already make a tentative theory.

പ്രയോജനപ്പെടുത്തൽ പ്രകാരം നിർമ്മാണ പ്രക്രിയയെ നോക്കി.

* It is an intelligent guess. അവയുടെ ഉപയോഗം ഉപയോഗിക്കുക.

* It is a gradual accumulation of indirect evidence.

അവയുടെ കാര്യങ്ങൾ ഉപയോഗിച്ച് നിർമ്മാണ പ്രക്രിയയെ നോക്കി.

* It may have numbers of predictions.

അവയുടെ കാര്യങ്ങൾ ഉപയോഗിച്ച് നിർമ്മാണ പ്രക്രിയയെ നോക്കി.

04) Controlled experiment : നിയന്ത്രിത പരീക്ഷണം

- * Test each prediction. ഓരോ പ്രവചനവും പരീക്ഷിക്കുക.
- * It is geared specifically towards rejecting one's favored hypothesis. ഇത് പ്രത്യേകമായി തള്ളേണ്ട പ്രവചനത്തെക്കുറിച്ചാണ്.
- * Often two experiments are conducted at the same time. അक्सോ ഒരേ സമയത്ത് രണ്ട് പരീക്ഷണങ്ങൾ നടത്താറുണ്ട്.
- * In one experiment all the variables are kept constant except one. ഒരു പരീക്ഷണത്തിൽ എല്ലാ വേരിയബിളുകളും സ്ഥിരമായിരിക്കും ഒഴികെ ഒന്ന്.
- * The other experiment - control experiment variable is left unaltered. മറ്റൊരു പരീക്ഷണം - കൺട്രോൾ പരീക്ഷണം വേരിയബിളിനെ മാറ്റാതെ.
- * The results of the two experiments are compared to determine if the tested variable has an effect on the outcome. രണ്ടു പരീക്ഷണങ്ങളുടെ ഫലം താരതമ്യം ചെയ്ത് പരീക്ഷിച്ച വേരിയബിളിന് ഫലപ്രസക്തമായ ഒരു ഫലം ഉണ്ടോ എന്ന്.

05) Theory - തത്വം.

- * When hypothesis is supported by evidence it becomes a theory. പ്രയോജനങ്ങൾക്കു താല്പര്യമുള്ളതായി തീർന്നാൽ അത് തത്വമായി മാറുന്നു.
- * More evidence, more acceptance confirms it. അതിനു താല്പര്യം, കൂടുതൽ സ്വീകാര്യതയ്ക്കു കാരണമാകുന്നു.
- * It is not a fact that cannot be changed. അത് തത്വമാണ് അത് മാറ്റാൻ കഴിയില്ല.
- * Ex: Theory of evolution - changed over time. ഉദാ: പരിണാമതത്വം - കാലക്രമേണമായി മാറുന്നു.

Role of Biology (ജീവ ജന്തുശാസ്ത്രത്തിന്റെ പങ്ക്)

Objectives of studying Biology.

ജീവ ജന്തുശാസ്ത്രം പഠിക്കുന്നതിന്റെ ലക്ഷ്യങ്ങൾ.

01) Understand ourselves better.

മനുഷ്യനെക്കുറിച്ച് കൂടുതൽ അറിയാൻ.

We study basic concepts related to structure and function of cells and organs. ശരീരത്തിലെ കോശങ്ങളുടെയും അവയുടെ പ്രവർത്തനങ്ങളുടെയും അടിസ്ഥാനപരമായ ആശയങ്ങൾ പഠിക്കുന്നു.

Growth, Reproduction, development.

ജന്മനം, പ്രജനനം, വളർച്ച.

Living things are capable of growing and reproducing
 ജീവികൾ വളരുകയും പ്രജനനം നടത്തുകയും ചെയ്യുന്നു.

They process hereditary molecules and are passed to
 offspring. മരുന്ന് പാരമ്പര്യ മൂലകങ്ങൾ പ്രോസസ്സ് ചെയ്ത് അവയെ അവരുടെ
 പുറം പെരുമാറ്റം വഴി കൈമാറുന്നു.

Homeostasis.

Capables of keeping internal environment constant
 ഉള്ളിൽ ഉൾക്കാമ്പ് സ്ഥിരമായി നിലനിർത്താൻ കഴിയുന്നു.

Evolution - ജീവശാസ്ത്രം.

With the time living things change according to the
 environment. കാലം കടന്നു പോകുമ്പോൾ ജീവികൾ അവയുടെ
 പരിസ്ഥിതിയോടനുസരിച്ച് മാറുന്നു.

The cell - As the basic unit of life,
(ശേഷു - ജീവന്റെ මූලික ඒකකය ලෙස)

Cell is the basic unit of structure and function of living things. ശേഷු යනු ජීවීන්ගේ ව්‍යුහයේ හා ක්‍රියාකාරීත්වයේ මූලික ඒකකයයි.

cells → Tissues → organs → Organ
(ශෛල) (සමාන) (අවයව) (ඔක්තීය)



Individual Systems
(නම් පද්ධතිය)

Cell theory (ශෛල න්‍යාය)

Proposed by schlieden and schwan in 1838.

It includes 3 principles. ඔබ්බේ 3 ක් ඇතුළත් වේ.

- All organisms are composed of one or more cells and metabolism and hereditary occur within these cells. සියලුම ජීවීන් ශෛල සකස් වේ හෝ වැඩිපමණක් සම්බන්ධ වන අතර ජීවමය හා ආරම්භක ලෙස ශෛල තුළ ඇති වේ.

- cells are the smallest living things, the basic units of organization. සෛල යනු ജീവജാലങ്ങളുടെ මූලික ඒකක වන කුඩාම ඒකකය.
- Cells arise only by division of previously existing cell. සෛල උග්‍රණයෙන්ම පමණක් ඇති වන්නේ පෙර පවතින සෛල බෙදීමෙන් පසුව.

Cell as a self-contained unit.

Cell though very small, it acts as an autonomous unit. සෛලය ඉතා කුඩා වුව ද එය ස්වයංපෝෂී ඒකකයක් ලෙස ක්‍රියා කරයි.

Cells ...

* carry out all vital biological processes.

සියලුම ජීව විද්‍යාත්මක ක්‍රියාවලි කරගෙන යාම දැක්වෙන්නේ.

* Oxidize the food molecules - produce energy

and store these energy rich molecules. දාහාර ද්‍රව්‍ය ඔක්සිකරණය කරන්න - ශක්තිය නිපදවා එය ගබඩා කරයි.

* build new structures and can replace worn out

cells by using nutrient molecules. භාහිර භාග්‍යාංග භාවිතයෙන් නව සෛල

නවීකරණය කරන අතර භාවිතයෙන් පිරිහී ඇති සෛල

සෛල ඉතිරි කර ගනී.

* repair and exchange gases with its surrounding.

දෘඩ වායුන් දូഷන්වැඩියා ശ්‍රී කිරීම හා ප්‍රමාදය කිරීම.

* replicate its own self

නමත්මම ඉතිරින්නය කරයි.

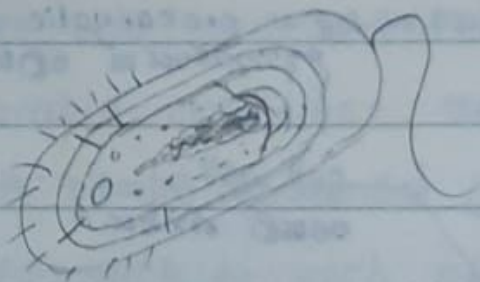
* have their own lifespan

මවුන්මේ දායකාමය දූත.

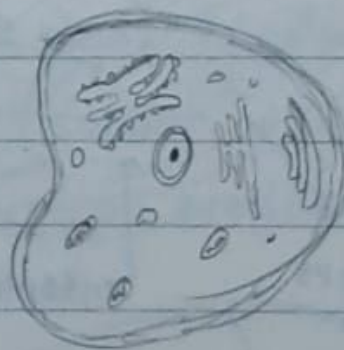
Prokaryotic and Eukaryotic cell

According to the cell structure there are two types of cells. කෙටු මනුෂ්‍යයට දූතම කෙටු මර්ග 2 තිමේ.

1. Prokaryotic cell



2. Eukaryotic cell

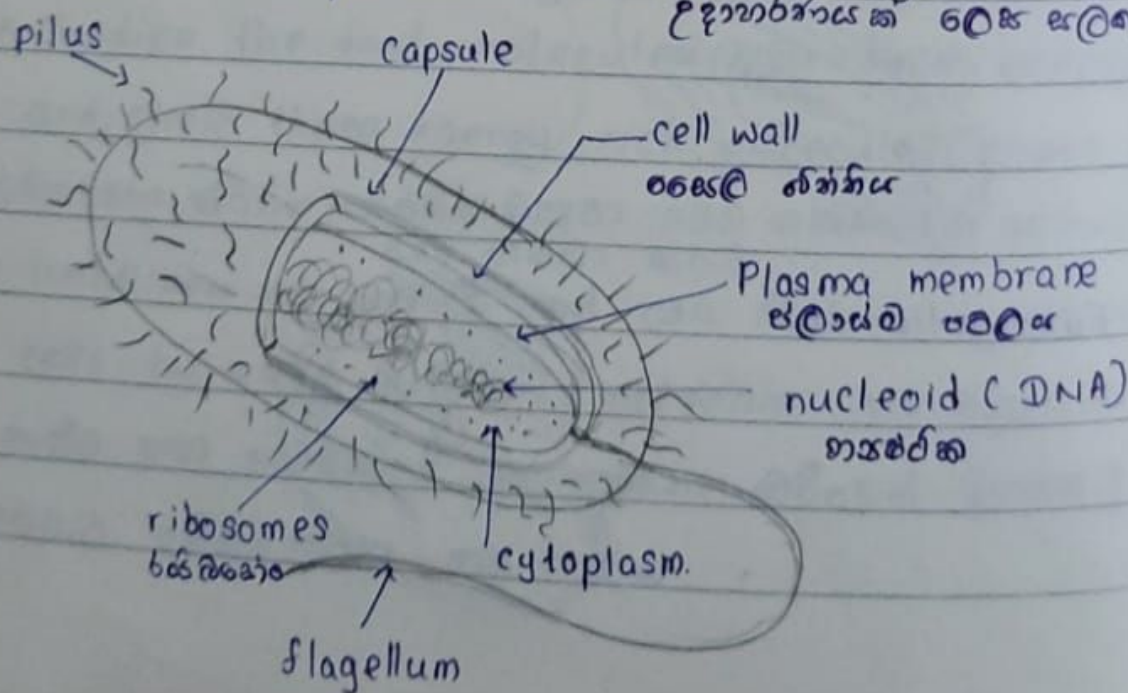


Prokaryotes is the term given to organisms without true nucleus. പ്രാപ്രകാരിക ജന്തു ജീവികൾക്ക് സത്യ സർവ്വകേന്ദ്രമില്ല.

hereditary material called DNA is not enclosed in a membrane / no membrane bounded internal structures. DNA ഇരിട്ട് ബന്ധിതമായ ജാർജർക്കു മൂലമുണ്ടാകുന്ന ജീവൻ / ജീവൻ ഇതിൽ ഉൾക്കൊള്ളുന്ന ജീവൻ

Structure of prokaryotic cell

We consider Bacterial cell as an example for Prokaryotic cell. දුර් කැක්ටීරියා කොටස prokaryotic කොටස ඇතුළත් වන්නාය. පිලස්, කැප්සියුල්, ප්ලාස්මිඩයන්, ටොක්සින.



Bacteria cell is composed of different parts.

බැක්ටീരിയා കോശങ്ങൾ വിവിധ ഭാഗങ്ങൾ ഉൾപ്പെടെയാണ്.

cell wall

Most bacteria are encased by a strong cell wall made up of peptidoglycon. බොහෝ බැක්ටීරියා peptidoglycon ഉൾപ്പെടെയാണ്. ഈ പദാർത്ഥം കോശത്തിന് വലിയ തරക്കം നൽകുന്നു.

No Eukaryotes possess cell walls with this type of chemical composition. ഈ രാസസംയോഗം ഉള്ള കോശങ്ങൾ eukaryotes ഇല്ല.

capsule.

The cell wall is covered by a capsule or slim layer. കോശഭിത്തിയെ മൂടുന്നതാണ് ക്യാപ്സ്യൂൾ. ഈ പദാർത്ഥം കോശത്തിന് അധിക സംരക്ഷണം നൽകുന്നു.

It gives additional protection to the cell from unfavorable conditions. ഈ പദാർത്ഥം കോശത്തിന് അനുകൂലമായ സാഹചര്യങ്ങളിൽ നിന്ന് സംരക്ഷണം നൽകുന്നു.

Helps to stick to each other and form colonies. കോശങ്ങൾ പരസ്പരം ചേർന്ന് കോളണി രൂപം നൽകുന്നു.

Helps to stick to the surface of the food supply or to the host cell. ഭക്ഷണസ്രോതസ്സിന്റെ ഉപരിതലത്തിലോ ആയിരുന്നോ കോശത്തിൽ ചേർന്ന് കോളണി രൂപം നൽകുന്നു.

Helps to stick to the surface of the food supply or to the host cell. ഭക്ഷണസ്രോതസ്സിന്റെ ഉപരിതലത്തിലോ ആയിരുന്നോ കോശത്തിൽ ചേർന്ന് കോളണി രൂപം നൽകുന്നു.

Plasma membrane.

Below the cell wall there is a membrane called plasma membrane. පහළින් කින්නිය පහළින් ප්ලාස්ම පෙරු තරින් පෙරු ය ය.

It is selectively permeable. It permits some molecules to pass through it. But not others.

සැසඳීමේදී පැහැදිලි කර ඇති පරිදි සැසඳීමේදී අනු භාවිතයෙන් කිසිදු
වෙනස්කමක් දක්නට නොමැත.

Cytoplasm

Plasma membrane encloses the cytoplasm.

ප්ලාස්ම පටලය cytoplasm දෘඪභාවය නිසා.

It looks granular due to the presence of ribosome.
 එයට හේതു වන්නේ නිසා අය කැරිති ලෙස පෙනේ.

There are enzymes, carbohydrates, lipids and genetic materials can be found in the cytoplasm.

cytoplasm രണ്ട് രണ്ടാ ന്ന നැമി.

Flagella

Many bacteria are equipped with one or more flagella. ക്വാഡ്രന്റ് ഡിസ്കീറ്റീഷ് flagella പക്ഷെ ബാക്റ്റീരിയ ഉൾപ്പെടെയുള്ള മറ്റ് ജീവികളിൽ ഇല്ലാത്തതാണ്.

It helps bacteria to move and to respond to stimuli.

ഇത് ബാക്ടീരിയ മുന്നേട്ടു പോകാനും പ്രതികരണത്തിനും സഹായിക്കുന്നു.

Eukaryotic cell.

Highly evolved cells possessing specialized organelles, a definite membrane bounded nucleus and highly organized cytoplasm. ഉയർന്നതരം കൂടിയുള്ള ജീവൻ ആണ് ഓർഗാനിസം. ഇത് പ്രത്യേകമായ ന്യൂക്ലിയസ് ഉള്ളതും പ്രത്യേകമായ സൈറ്റോപ്ലാസ്മ ഉള്ളതും ആണ്.

great variations in shapes, size and type.

හැඟුයේ ප්‍රධානම හා මු්තමයනි නිමාල බෙහෙසිකර්.

Jelly like translucent protoplasm bound by a plasma membrane. ഇറാലറി പരിവൃത്തിൽ മുള്ളു മുൻ പാർക്കാക്കു ജ്യോരറി ഇറാലിഡ ചേറ്റി രെസ്.

protoplasm - 2 parts.

cytoplasm and nucleus.

Cytoplasm - number of compartments.
 തുല്യ ഘാത.

called as organelles and carryout different functions. ഊർജ്ജങ്ങൾ നൽകുന്നതിനും മറ്റ് ജീവികാർയങ്ങൾ ചെയ്യുവാൻ.

These organelles may or may not be bound by a membrane. Ex.

Mitochondria, Ribosome, Golgi apparatus, Lysosome
 തുല്യ ജീവികാർയങ്ങൾ ചെയ്യുവാൻ തോന്നിയെ ഉപയോഗിക്കുന്നു.

Types of Eukaryotic cells

plant cells

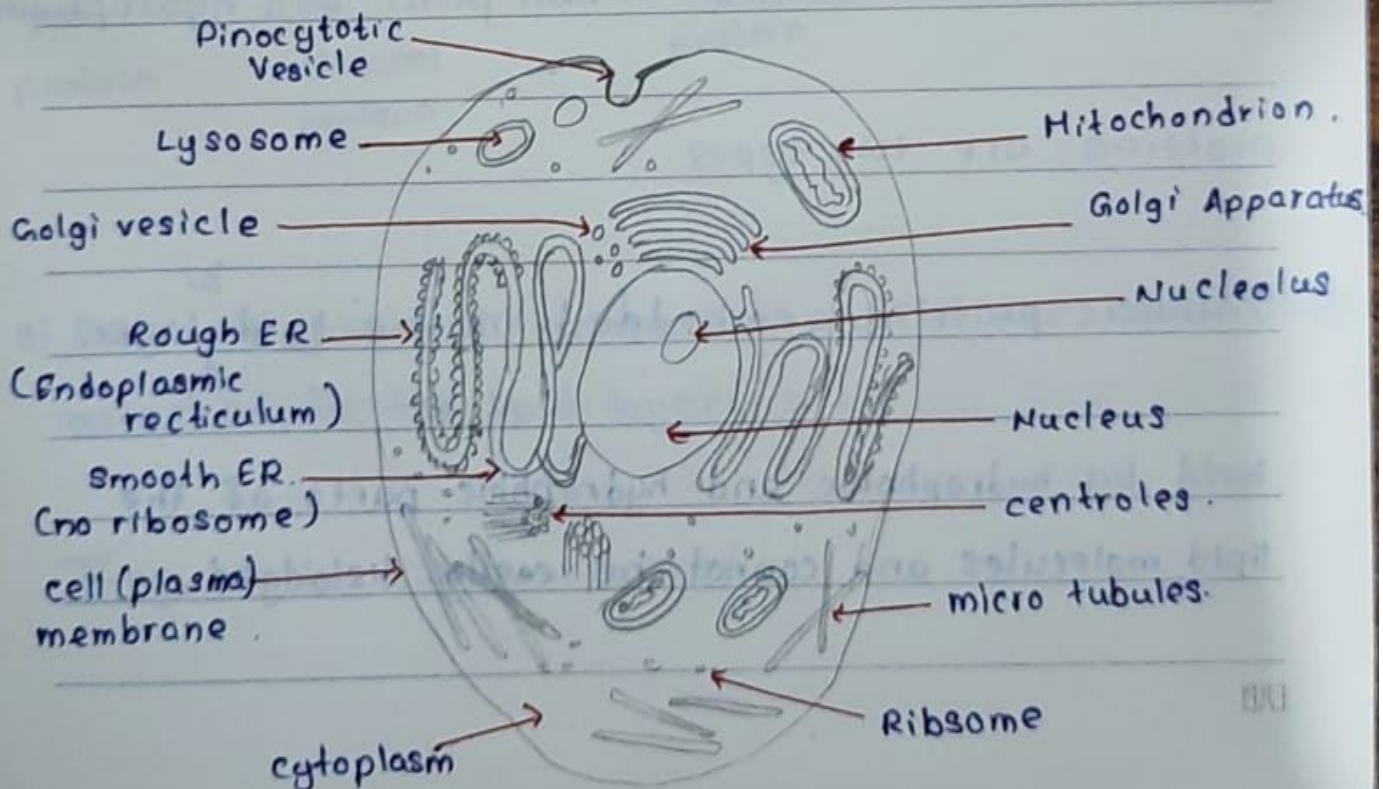
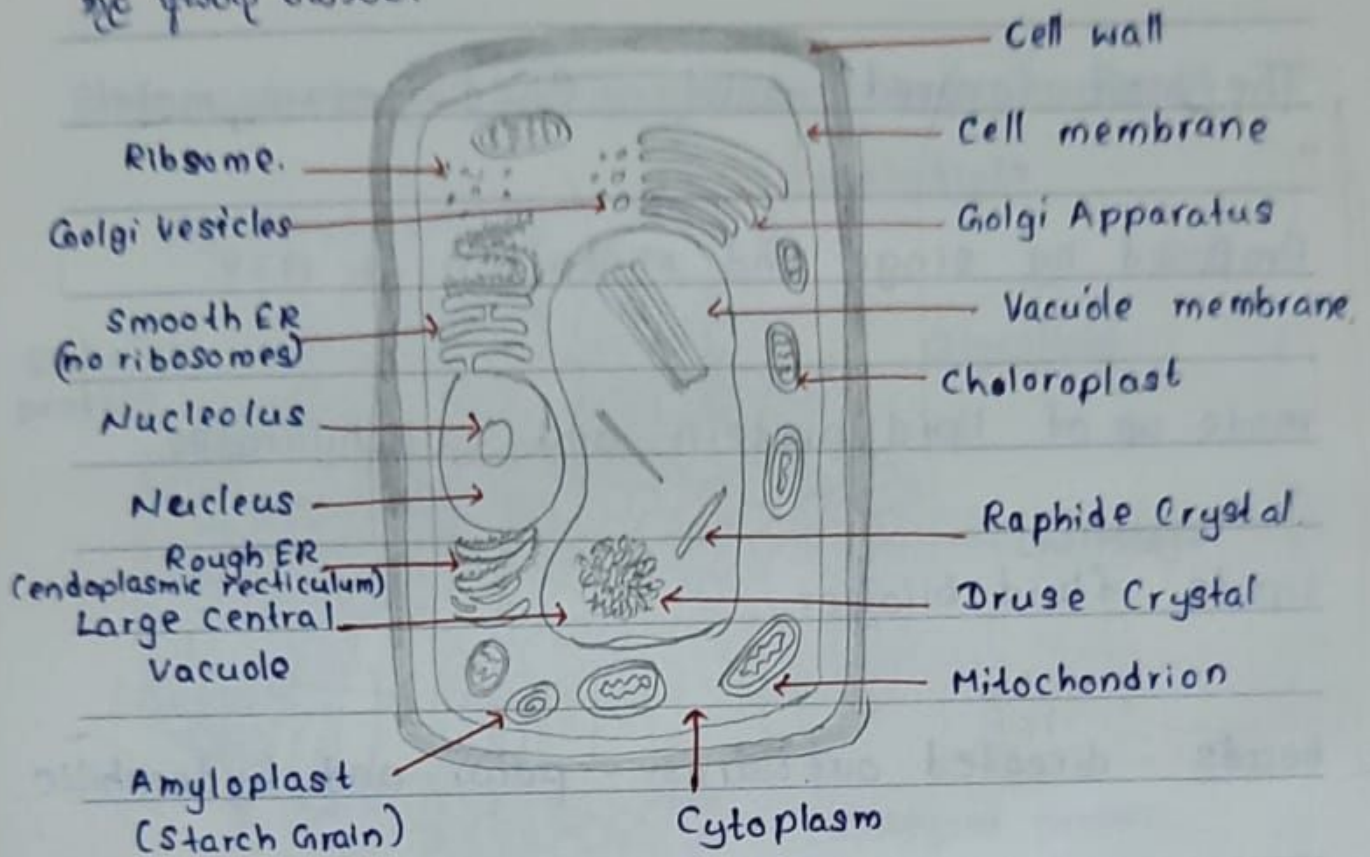
animal cells.

Major differences are (പ്രധാന വ്യത്യാസങ്ങൾ)

1. Plant cell contains a cell wall but animal cell does not. അതായത് രണ്ടും തുല്യ വ്യത്യാസങ്ങൾ ഉണ്ട്.

2. Plant cell contains a large central vacuole but animal cell contains number of small vacuoles. അതായത് രണ്ടും തുല്യ വ്യത്യാസങ്ങൾ ഉണ്ട്.

3. Plant cell contains a chloroplast but animal cell does not. ~~animal cell does not contain chloroplast~~ chloroplast found only in plant cell.



Structure of the Plasma membrane.

(ക്രമീയ പരമാവർത്തമാനം)

The most favored model - fluid mosaic model.

Proposed by singe and Nicholson in 1972.

made up of lipid, protein and carbohydrates.

Lipids - fluid bilayer.

heads - directed outwards - polar and hydrophilic.

Tails - directed inwards - non polar and hydrophobic.

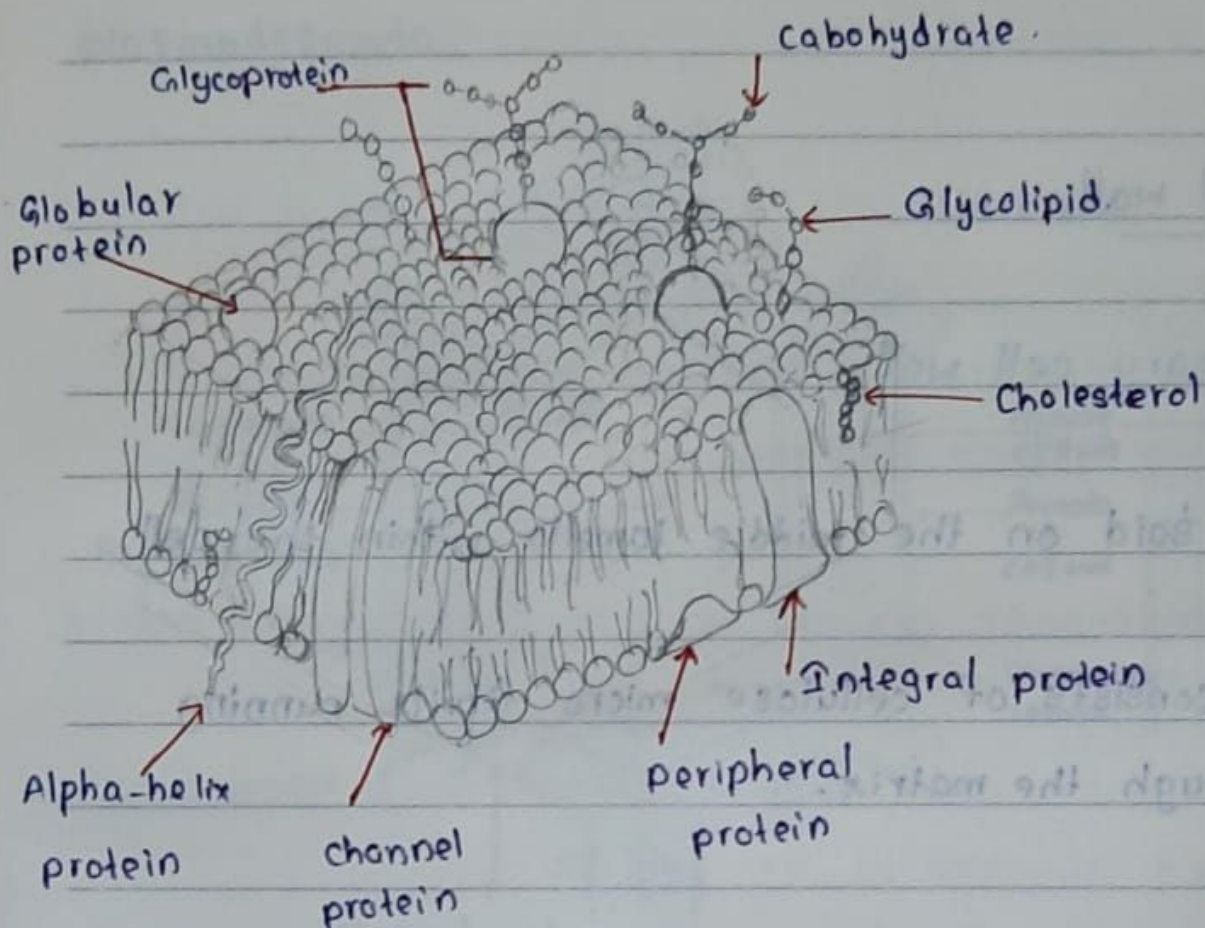
proteins are two types.

i) Intrinsic proteins - embedded in the lipid layer.

held by hydrophobic and hydrophilic parts of the lipid molecules and cannot be easily dislodged.

Some protein molecules are small and only a part projects at the surface.

Some are large and projects at both surfaces.



2) Extrinsic proteins / peripheral protein are superficially attached to the lipid layer.

They can be easily removed.

Carbohydrates forms the antennae projecting on the surface.

The cells recognize one another by these.

Cell wall

Primary cell wall.

Is laid on the middle lamella within the cell.

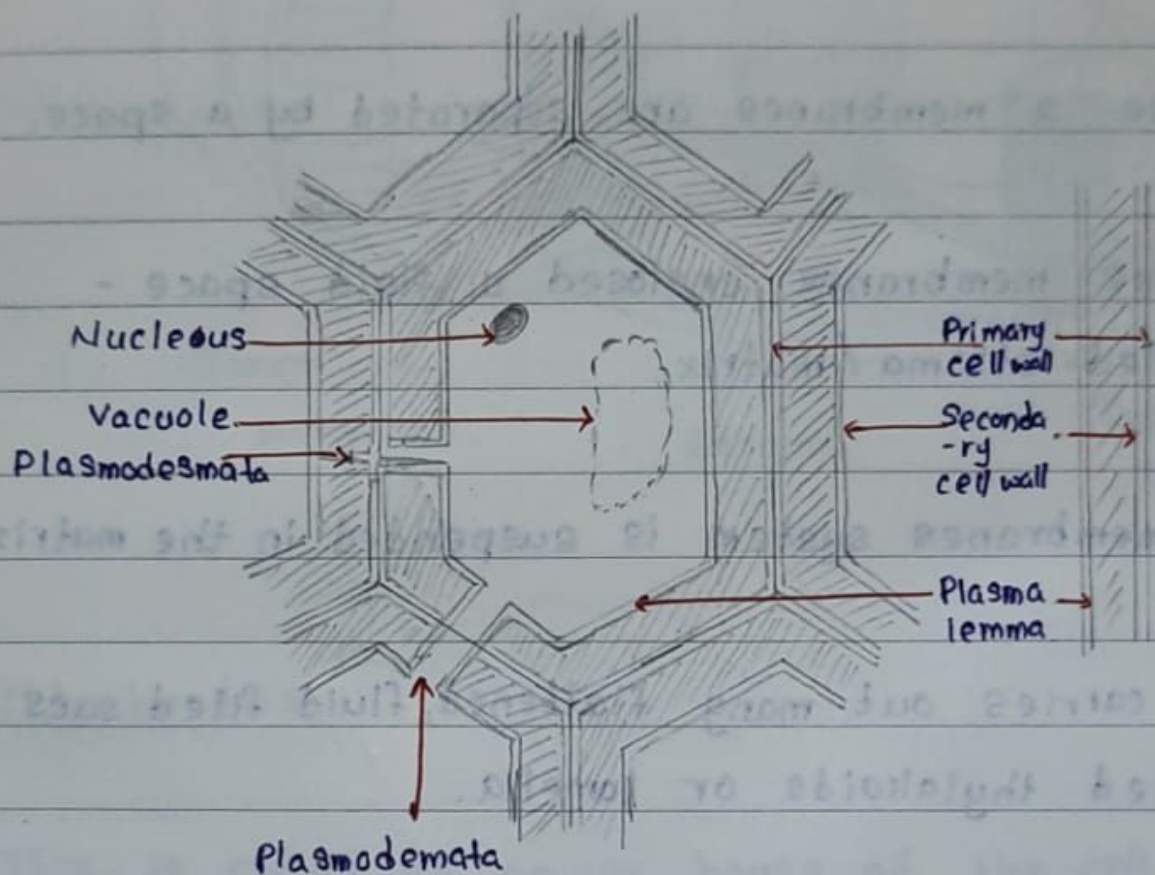
It consists of cellulose micro fibrils running through the matrix.

Secondary cell wall.

It is built up by addition of cellulose layers on the inside surface of primary cell wall.

Plasmodesmata

The cell wall does not form a continuous layer and is interrupted by minute pores called plasmodesmata.



Arrangement of various layers of cell wall in a mature plant cell.

Cellular Organellas

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Chloroplast

It is surrounded by 2 membranes.

These 2 membranes are separated by a space.

These membranes enclosed a fluid space - called stroma / matrix.

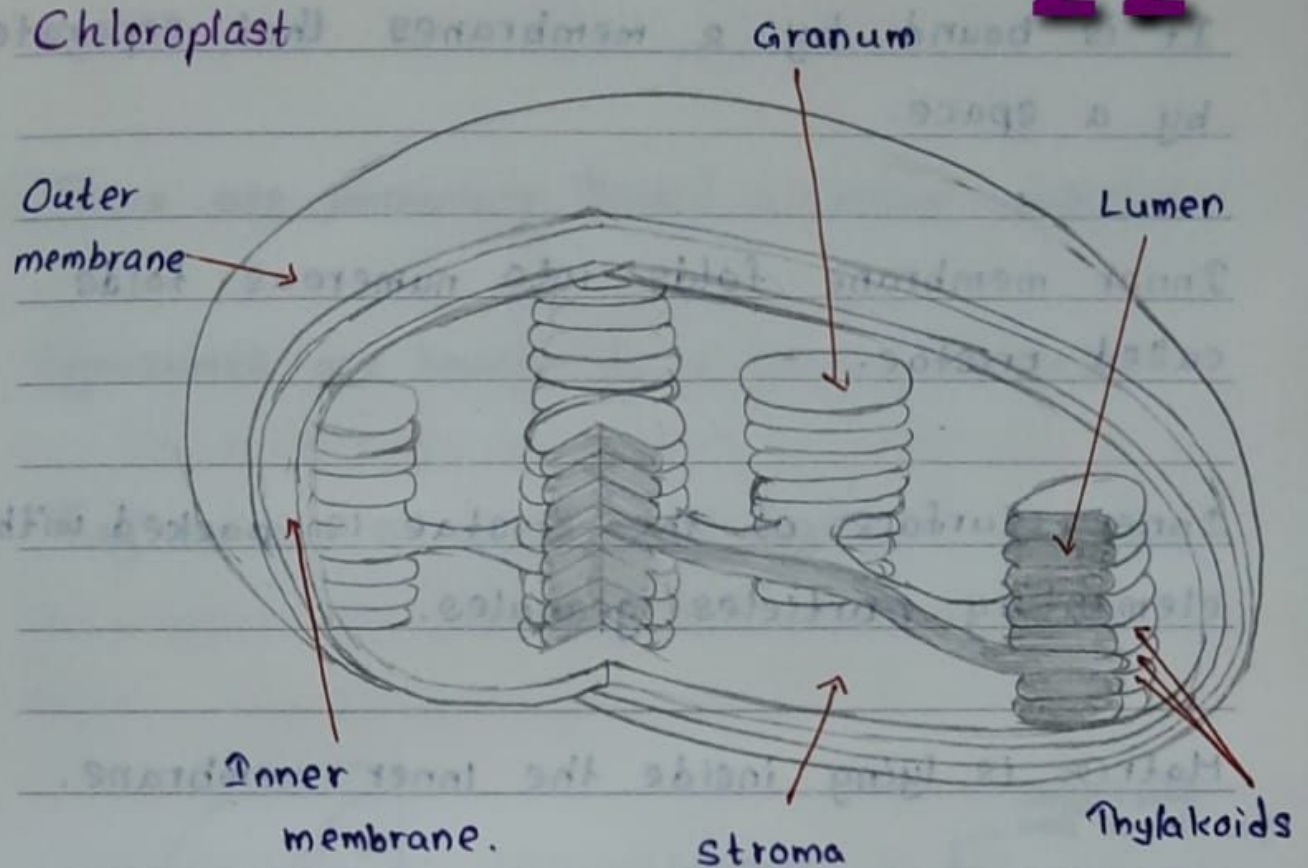
A membranes system is suspended in the matrix.

It carries out many flattened, fluid filled sacs called thylakoids or lamella.

These thylakoids stack to form grana.

Various grana are jointed by integrated lamella.

Chloroplast



Mitochondria.

This is called as power house of the cell.

It is specialize in the synthesise of ATP, the usable chemical energy.

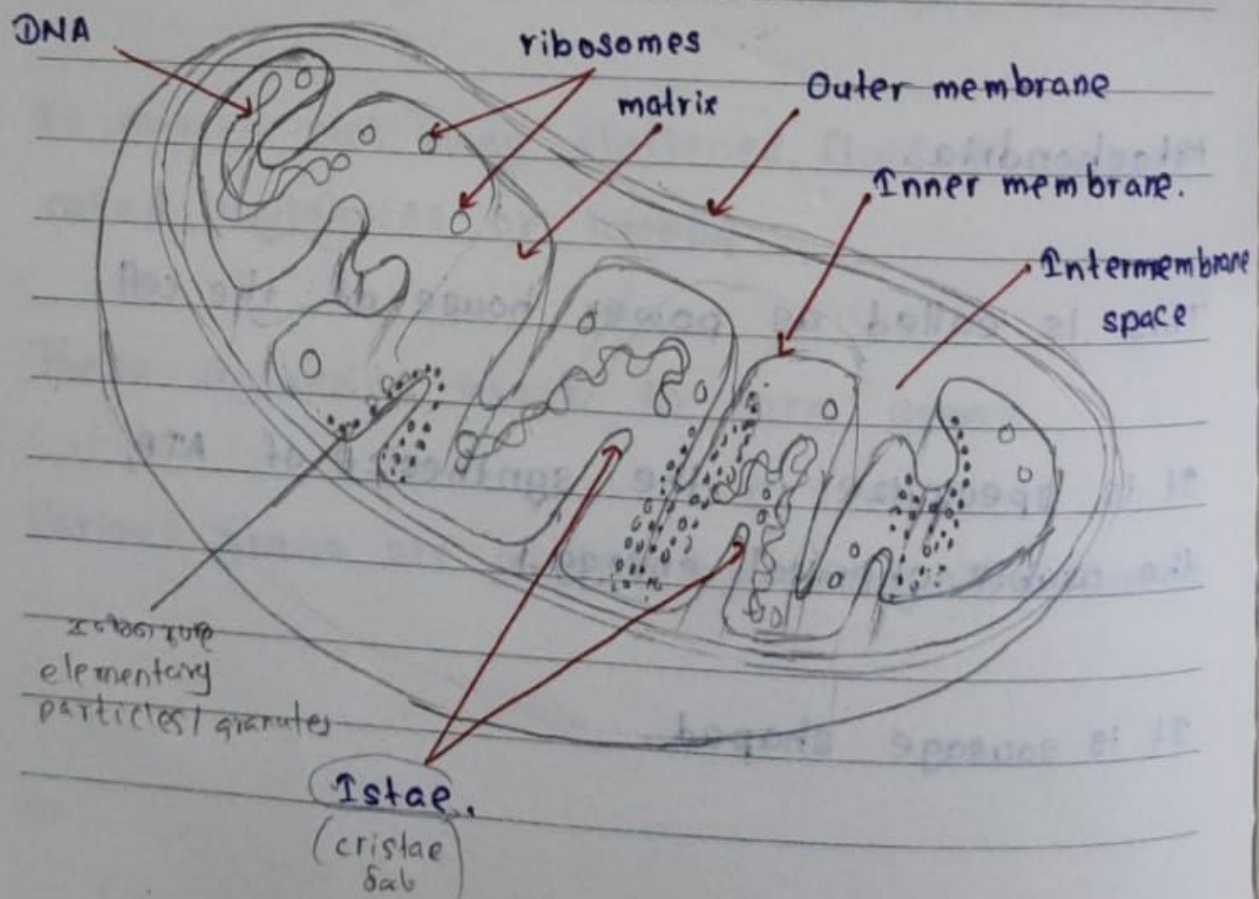
It is sausage shaped.

It is bound by 2 membranes that separated by a space.

Inner membrane folds into numerous folds called cristae.

Inner surface of the cristae is packed with elementary particles / granules.

Matrix is lying inside the inner membrane.

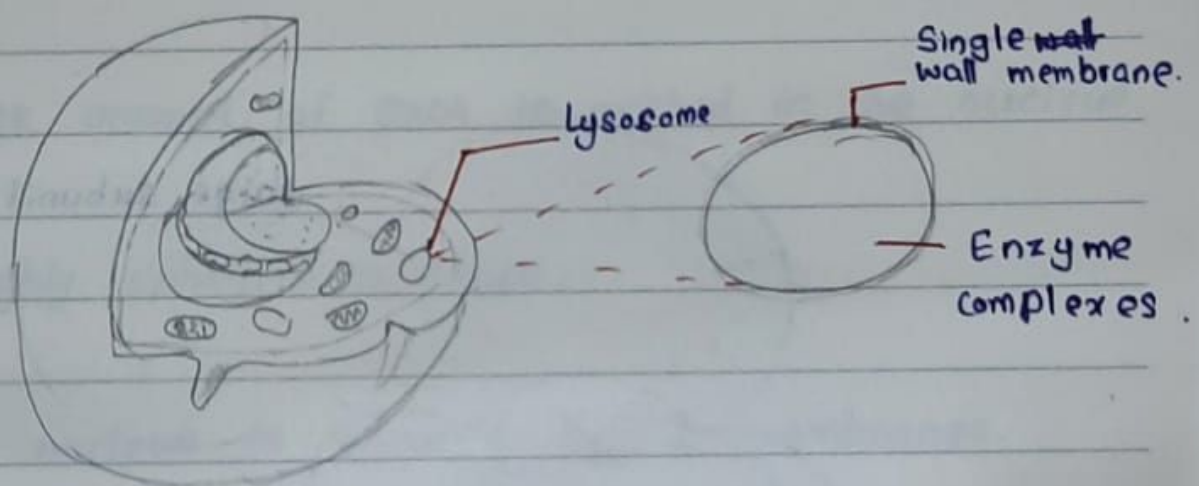


Lysosome.

There are membrane bound digestive vesicles.

Lysosomes are bound by a single membrane.

They contain a mixture of digestive enzymes.

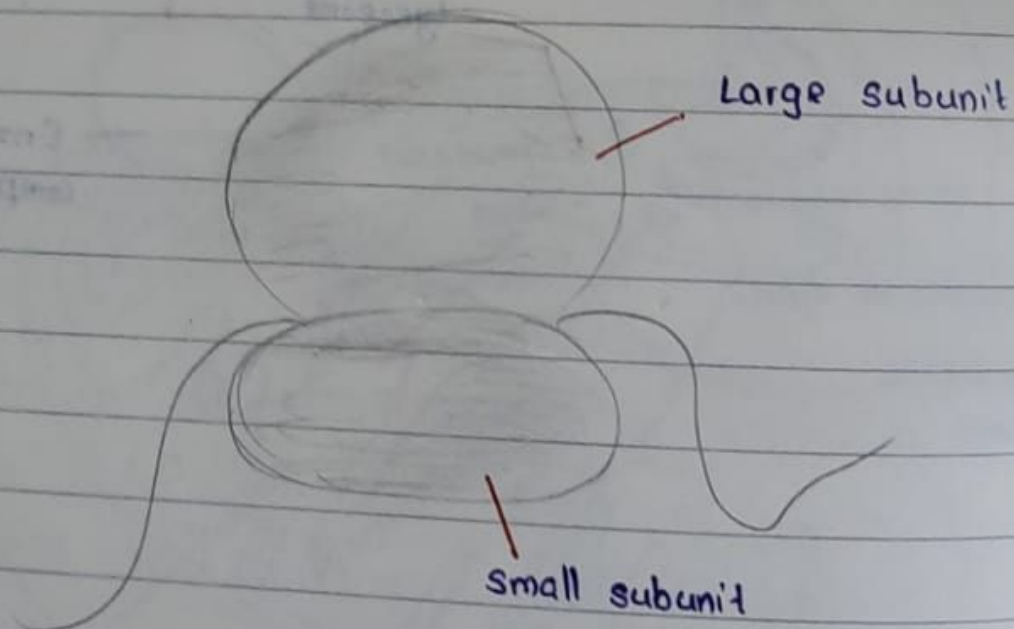


Ribosome.

They are spherical bodies found in the cytoplasm of the cell.

It is composed of large and small subunit.

They serve as sites of protein synthesis.



Instructions to make a protein
(written on a mRNA molecule in
the nucleus)

Nucleus

This is the largest and most easily seen organelle.

This is the core of the cell which command all metabolic activities.

In animal cells, nucleus is located in the central region.

Large amount of DNA is packed in the nucleus.

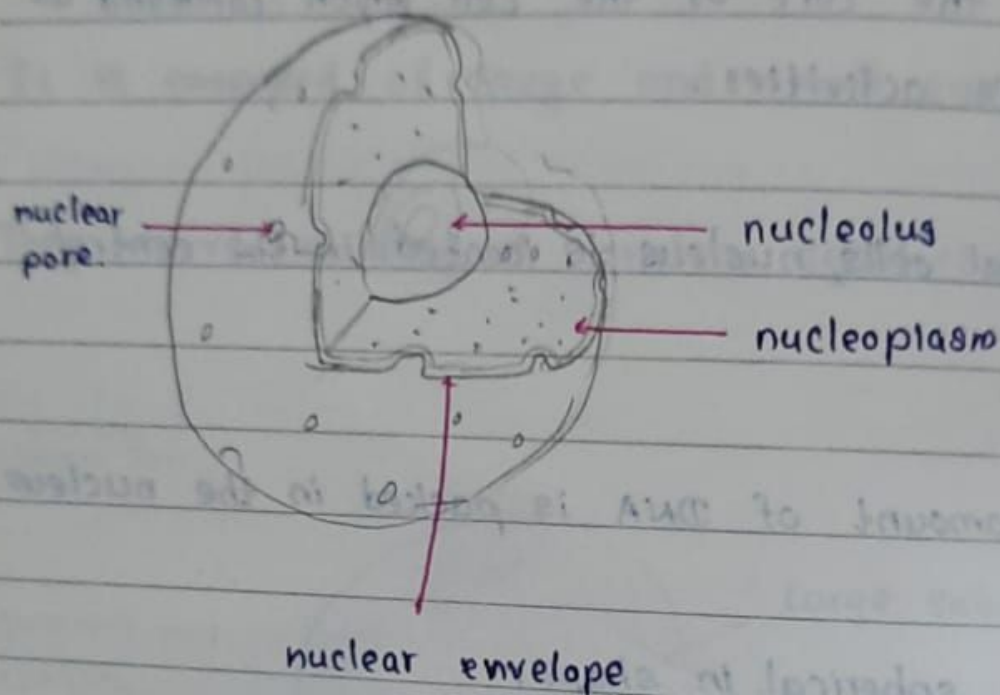
Roughly spherical in shape.

The nucleus is Encased by 2 membranes.

Nucleolus is the site with in the nucleus where ribosome are made.

It contain large amount of DNA and RNA.

The space between the nuclear envelope and the nucleolus is filled by the transparent, granular nucleoplasm.

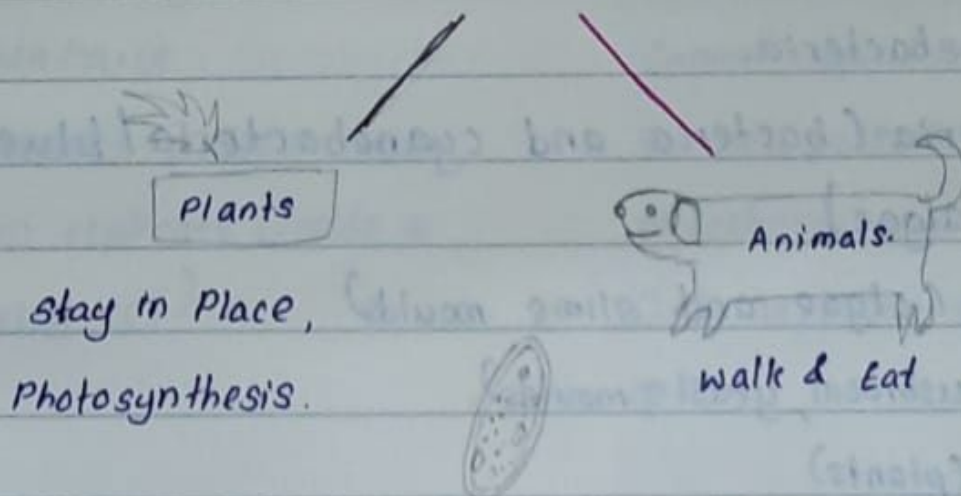


Classification of Life.

In early history people knew only plants and animals.

Linnaeus developed his system of classification, there were only two kingdoms, Plants and Animals

Life traditional.



The invention of the microscope led to the discovery of new organisms and the identification of differences in cells.

So, the two-kingdom system was no longer useful.

Today the system of classification includes six kingdoms.

The six kingdoms of life.

1. Archaeobacteria.
2. Eubacteria (bacteria and cyanobacteria/blue green algae).
3. Protists (algae and slime moulds)
4. Fungi (mushroom, yeasts, moulds)
5. Plantae (plants)
6. Animalia (animals)

In addition to this, there are, 'viruses'.

These are sub-microscopic agents that can be seen only through the electron microscope.

They are unable to grow and reproduce outside living cells.

They share both living and non living characteristics.

Living and non living characteristics of viruses.

Living Non-Living

Possess genetic material

Non cellular.

DNA/RNA

Cannot reproduce by themselves

Proteins protein coat

Do not possess ribosomes,

Can replicate (inside a living cell)

mitochondria etc.

Only have one type of nucleic acid.

Kingdom 1. Archaeobacteria.

These are a diverse group of bacteria.

Archaeobacteria are different from Eubacteria, from the places where they live.

They live in extreme conditions which are not suitable for other normal life forms.

There are three major known groups of Archaeobacteria.

1. Methanogens.

2. Halophiles.

3. Thermophiles.

1) Methanogens.

They live in oxygen less environments.

They are found in sewage treatment plants, bogs, and the intestinal tracts of some animals.

2) Halophiles

Are bacteria that thrive in high salt concentrations such as sea water.

3) Thermophiles.

Are the heat-loving bacteria found in hot water springs.

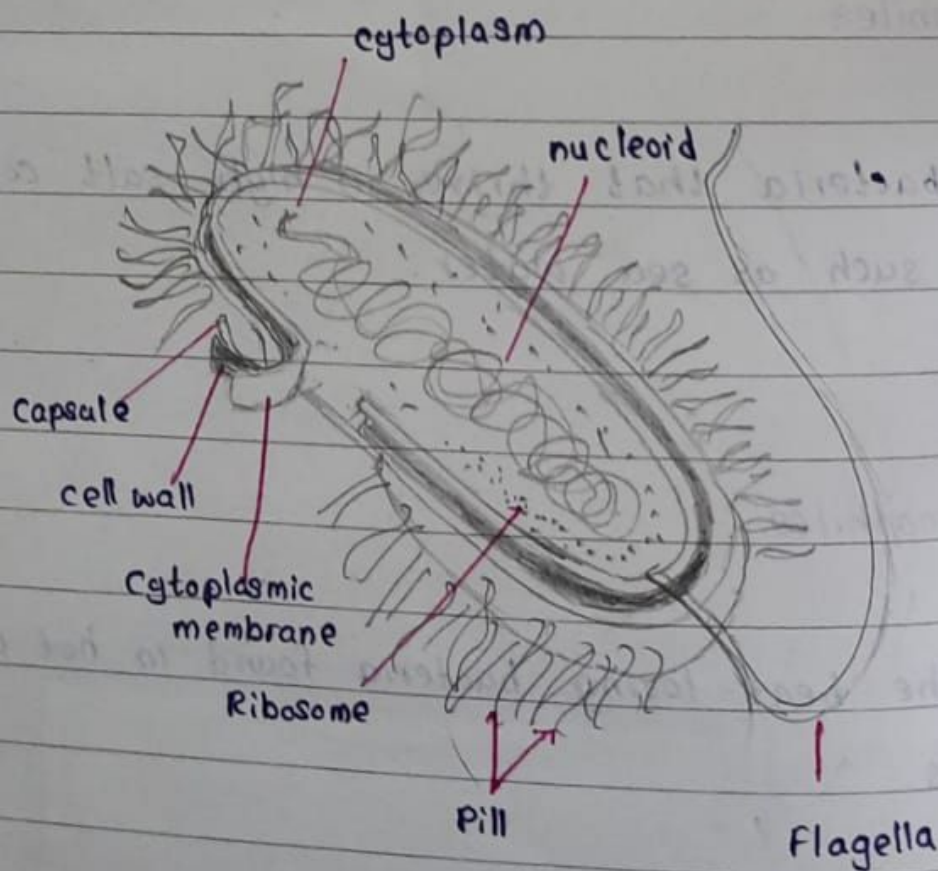


Kingdom 2. Eubacteria.

Includes Bacteria and Cyanobacteria.

Bacteria

Tiny minute organisms that can not be seen with the naked eye.



These are prokaryotic, photosynthetic (can make their own food) simple life forms.

Ex: Microcystis.

Kindom 3; Protista (Algae & slim moulds)

All are Eukaryotes.

All algae contain chlorophyll a

Depending on the type of algae along with chlorophyll 'a' they can have 'chlorophyll b', 'c', 'd', or 'e'

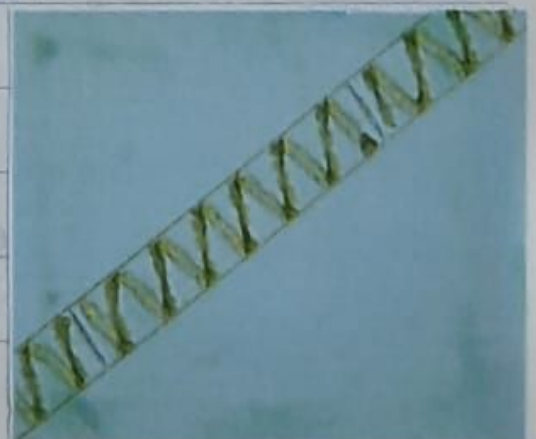
Thus these algae are all photosynthetic protista.

Algae may occur in oceans, lakes, tanks, rivers streams and also in damp soil and on bodies of other organisms.

Ex. Chlamydomonas, Spirogyra.



spirogyra.



Kindom 4 : Fungi

Non-photosynthetic (no chlorophyll so can't produce their own food), eukaryotic organisms,

Heterotrophic (get their food from others)

Most are small, microscopic, filamentous branched forms that bear spores.

Ex: mushrooms



Kindom 5 : Plantae.

Two types of plants as vascular or non-vascular.

Non-vascular plants are more primitive.

Ex: Thalloid liverworts.

Mosses

ferns.

vascular plants

can be of two types;

Vascular plants.

Can be of two types,

1. Open seeded plants (ex: Cycas.

2. Enclosed seeded plants.

Can be

Dicot plants

Monocot plants.

Differences between dicot and monocot plants.

Monocots				
				
One cotyledon	Veins usually parallel	Vascular bundles usually complexly arranged	Fibrous root system	Floral parts usually in multiples of three
Embryos	Leaf venation	Stems	Roots	Flowers
Dicots				
				
Two cotyledons	Veins usually netlike	Vascular bundles usually arranged in ring	Taproot usually present	Floral parts usually in multiples of four or five

Kingdom 6 : Animalia.

This kingdom consists of

multicellular (composed of more than one cell)
heterotrophic (feed on other organisms - can't produce their own food) organisms.

Divided into 2 main Groups.

1. Invertebrates.
2. Vertebrates.

Invertebrates.

Consist of

- Porifera - Sponges.
- Echinodermata.
- Cnidaria.
- Plathelminthes
- Nematoda.
- Annelida
- Mollusca
- Arthropoda.

Vertebrates.

Consists of.

- Pisces - Fish.
- Amphibia.
- Reptilia.
- Aves - Birds.
- Mammalia.

Phylum : Porifera.

Called pore bearers as their body wall contain minute pores.

All are aquatic marine or fresh water.



Phylum Cnidaria.

Radial symmetrical animals.

They have specialized cells called Cnidocytes
- "stinging cells"



Phylum Platyhelminthes (flatworms).

Soft, dorso-ventrally flattened animals with
bilateral symmetry.

Most are Parasitic.



Phylum Nematoda.

Commonly known as round worms.

Occur almost every where in soil, fresh water, salt water and as parasites in the body.

Body wall is covered by a tough cuticle.



Phylum Annelida.

Elongated segmented body.

Body is covered by a transparent cuticle.



Phylum Mollusca.

Live in fresh water, marine water and on land.

Soft bodied.

Have a muscular foot used for locomotion.

Ex: snail.



Phylum Arthropoda.

Largest group of animals.

Reasons for the success of Arthropods:

a) Chitinous exoskeleton.

They have a unique chitinous (made up of chitine) cuticle which is light weight and tough.

It protects the animals from predators.

It is imprevious to water and so protects the body from drying and has made the arthropods survive on land in almost all types of environments.

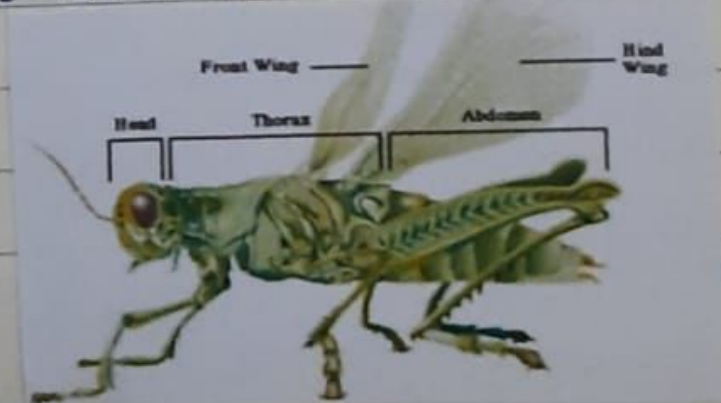
Insects have chitinous wings that help them to fly.

The biting mouth parts, lenses for the eye, and sensory organs are also made up of chitin.



2) Segmentation

Body is segmented into distinct regions as head, thorax and abdomen; each one is specialized to perform a specific function.

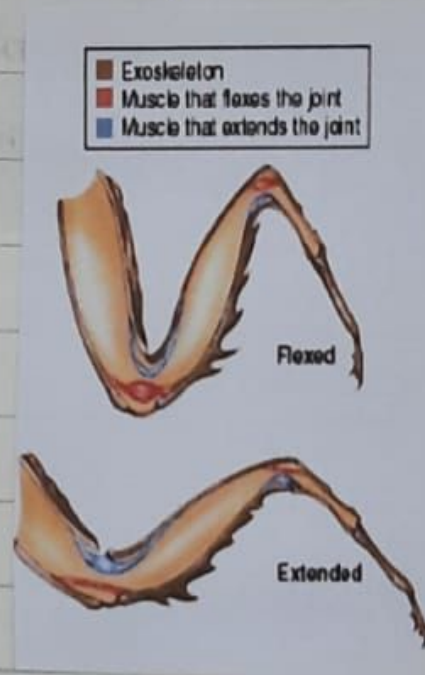


3) Jointed appendages

The appendages have been modified to perform different functions.

It increases the efficiency and complexity of the activity.

It is also help the arthropods to adjust to different habitats.



4) Ecdysis / molting

The chitinous skeleton does not allow the body to grow and hence to overcome it, ecdysis has evolved.

The animal shed off the exoskeleton periodically and grows larger and secretes the new exoskeleton.

phylum Echinodermata.

All are marine animals.

Spiny skinned animals. The spines fixed or movable.

Radial symmetrical animals with a water vascular system.

Have a anterior - posterior flattened body.



Phylum Chordata.

Class - Pisces (Fish)

Two types of fish.

* Cartilaginous fish and Bony fish.

Cartilaginous fish.

Cartilaginous fish have cartilaginous skeleton.

They do not have a swim bladder so that they must keep swimming.

They are viviparous - live bearers.

Ex: Rays and Sharks.



Bony fishes.

Have bony skeleton.

Have spindle shaped body.

Respire through gills.

Have scales in the body.

Have a swim bladder help them to rise and sink.

Have two chambered heart.

Ex : Rainbow Trout



Class - Amphibia.

They live in water and on land

They have smooth and moist skin.

No scales present.

They have two pairs of limbs.

Their heart is three chambered.

Ex: Tree frog.



Class - Reptilia

They have dry horny scaly skin.

They have two pairs of limbs.

Their heart is three chambered.

Ex: python.



Class - Aves.

Distinct feature of birds is flight.

Characteristic features that help the birds in flight.

They have a stream lined body covered by feathers to help to reduce the resistance.

The fore limbs are modified into wings. They are also covered by feathers.

Have light, porous bones with air cavities to reduce the weight.

Have a strong breast bone called keel to which the flight muscles are attached.

Heart is four chambered which allows complete separation of the oxygenated blood from the deoxygenated blood.

~~Have large.~~

Hind limbs are adapted for perching.

The tail is used for changing the direction.

Have a constant body temperature called warm blooded.

They maintain a high ~~main~~ metabolic rate in order to lift off the body from the ground.

The heart of the ~~body~~ birds beats faster
in order to supply large amount of oxygenated
blood into the body.

Have large eyes and strong sense of sight.

The field of vision is also great.

They have a remarkable ability to judge the
distance from a close range as well as from
a great height.

The disappearance of right ~~away~~ ovary and
right oviduct reduces weight and is an
adaption



Class - Mammalia.

They are viviparous and nourish the young ones with milk secreted from mammary glands.

Body is covered with hair.

External ear present.

Spiny ant eater is an exceptional egg laying mammal.

Remarkable mammals.

Spiny ant eater is an egg laying mammal.

In kangaroo, young ones are 3 cm long when born.

They crawl into the pouch and feed on the milk.

Shrews are the smallest, insectivorous mammals.

Bats are flying mammals. Their fore limbs are modified into wings.

Whales, Dolphins are aquatic mammals. Fore limbs are modified into flippers.

Man, Monkey, Gorilla are highly evolved mammals.

They have opposable thumb. Intelligent animals.

Mammals



Communities And Ecosystems.

What is "Community"?

All the organisms that live together in a place is called a community.



What is "co-evolution"

The plants, animals, protists, fungi bacteria etc. live together and have changed and adjusted to one another continually over a period of millions of years.

Such interactions are called coevolution.

Symbiotic relationships.

Are shown by two or more kinds of organisms live in a place in permanent relationships.

Kinds of symbiosis.

Commensalisms.

Mutualism.

Parasitism.

Commensalism.

This relationship gives benefits for one species and neither hurts nor helps the other.

Individuals of one species is physically attached to members of another.

Ex. Epiphytes - They are the plants that grow on the branches of other plants.

Host plant is unharmed and the epiphyte benefits by having the substrate for growing.

ex. orchid.



Mutualism.

54

In this relationship both species benefits.

ex: Ants and aphids.

Aphids are small insects that suck fluids from the phloem of living plants with their piercing mouth parts.

They excrete them through their anus.

Ants carry the aphids to new plants and then consume the excreted materials of aphids as food.

So, Aphids get new plants and ants get food.



Parasitism.

55

↑

The parasite is much smaller than the prey that closely associated with it. This relationship is harmful to the prey organism and beneficial to the parasite.

Ex. Lice

mites.

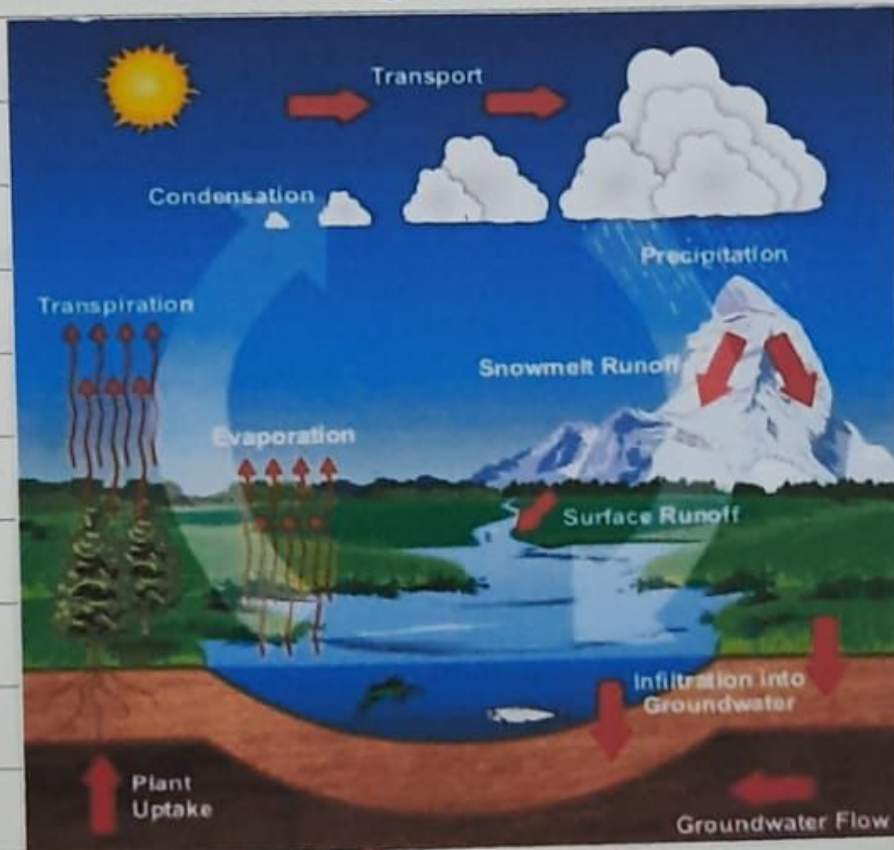


Dynamics of the Ecosystem

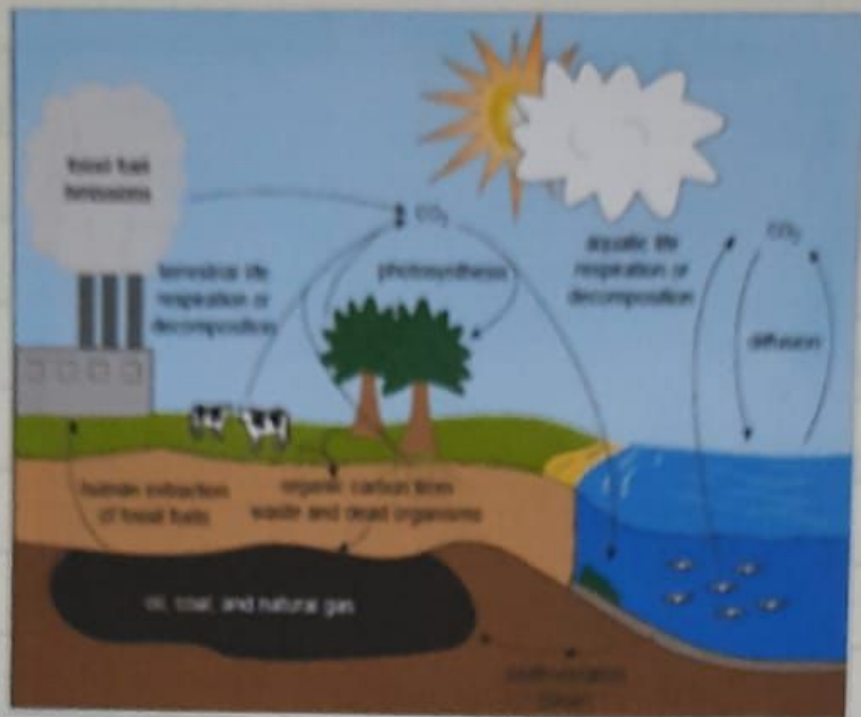
every ecosystem has regulated transfer of energy and a controlled cycling of nutrients.

All of the substances cycle through ecosystems in biogeochemical cycles.

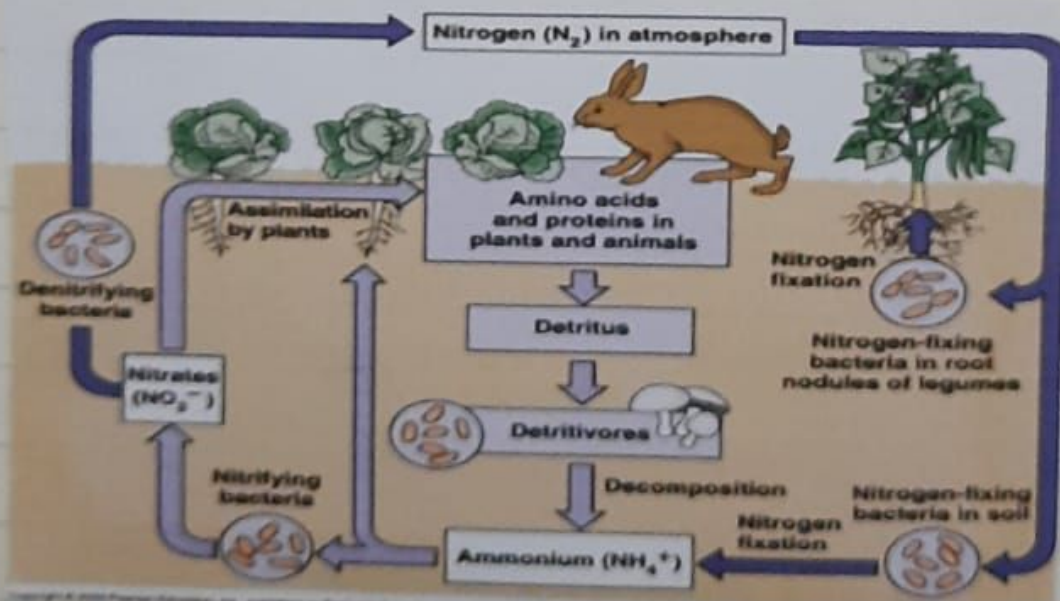
Water cycle



Carbon Cycle



Nitrogen Cycle



Trophic levels.

An ecosystem is composed of autotrophs and heterotrophs.

Autotrophs are the plants, algae and some bacteria that are able to capture light energy and manufacture their own food.

They are primary producers.

Heterotrophs are the animals, fungi and most other organisms, who obtain organic molecules that have been synthesized by autotrophs.

They are consumers.

There are several different levels of consumers.

Primary consumers are herbivores.

They feed directly on the green plants.

Secondary consumers are carnivores and parasites of the animals feed in and on the herbivores.

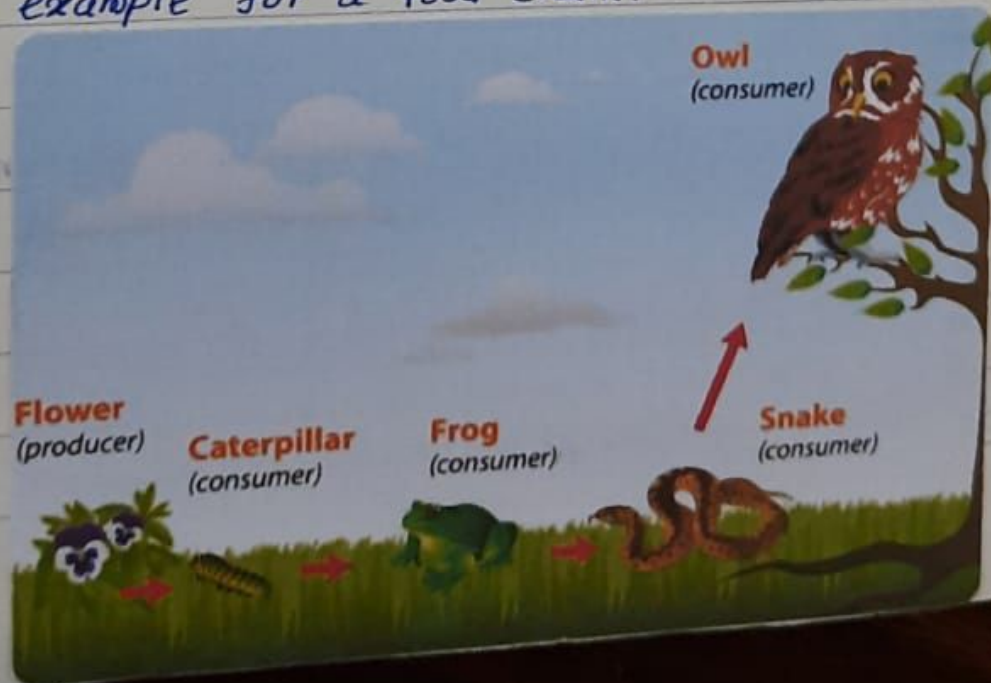
Decomposers breakdown the organic matter accumulated in the bodies of other organisms.

All of these categories are occurred in ecosystem.

They represent different "Trophic Level"

Organisms from each topic level feed on one another make up a series called "food chain"

An example for a food chain.



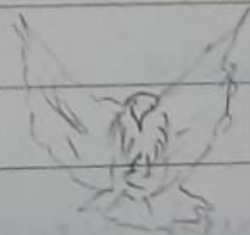
In real life it is rare for a given kind of organism to feed only on one other type of organism.

Usually each organism feed on 2 or more kinds and it is eaten by several other kinds of organisms.

When diagrammed it appears as a series of branching lines, than a straight line.

It is called a food web.

An exaple for a food web.



BIOMS

61

Bioms is the largest terrestrial community or the largest ecological unit.

It has specific kinds of plants and animals.

It has specific distinct climatic conditions.

Biomes are ~~des~~ distributed on earth as broad belts.

The major biomes.

1. Tundra

Found across Asia, Europe and North America

Rainfall below 25 cm.

Poor Light and extremely low temperature.

Snowfall is scanty.

spare vegetation.

Dwarf willow trees, lichens, mosses are found

s. Taiga.

It is occurred at the east west band across North America, Europe and Asia

Rainfall 10-35 cm.

Average temperature 6°C in winter and 20°C in summer.

Evergreen trees



3. Deciduous forest.

Occurred in temperate region of North Central Europe, Eastern Asia and Eastern United States.

Rainfall 75-105 cm.

Broadleaved hardwood trees are found.



4. Tropical rain forest.

Occurred in equatorial regions.

Rainfall exceeds 140 cm.

Warm and humid climate.

Luxuriant and varied vegetation.



6. Chaparral.

Extends along Mediterranean sea, Pacific coast of South America, Chile, South Africa and South Australia. Limited winter rain followed by drought

Vegetation composed of broad-leaved evergreen shrubs, bushes, and small trees usually less than 2.5 m tall; together they often form dense thickets

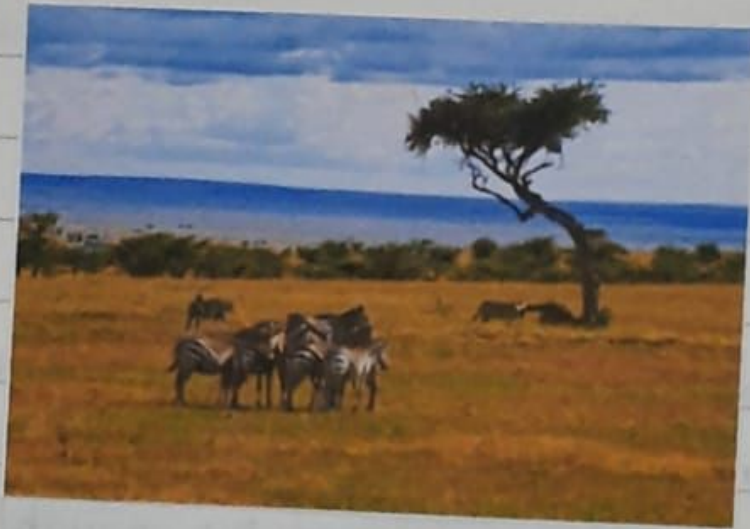


6. Tropical savannah

South America, Africa, Asia, and Australia.

Warm climate

Rainfall 100-150 cm



7. Grasslands

Prairies of Canada and USA, Pampas of south America, Steppes of Europe and Asia, Veldts of Africa.

Rainfall is about 25-75 cm.

Short and tall grasses found.



e. Desserts

South western USA, Mexico, coastal area of Chile and Peru, North Africa Sahara dessert

Extremely low rainfall. Less than 25 cm, high evaporation.

Hardy grasses found.



Nutrition

Autotrophic and Heterotrophic Nutrition.

Living organisms obtain chemical energy, they need to perform activities, in one of two ways.

Autotrophic -

Autotrophs store chemical energy in carbohydrate food molecules they build themselves.

Food is chemical energy stored in organic molecules.

Food provides both the energy to do work and the carbon to build bodies.

Because most autotrophs transform sunlight to make food, we call the process they use photosynthesis.

Only three groups of organisms - plants, algae and some bacteria. - are capable of this life-giving energy transformation.

Autotrophs make food for their own use, but they make enough to support other life as well.

Almost all other organisms depend on these three groups for the food they produce.

The producers, as autotrophs are also known, ~~being~~ begin food chains which feed all life.

(a) Plants.

(b) Algae.

(c) Photosynthetic bacteria.

Heterotrophs cannot make ~~either~~ their own food, so they must eat or absorb it.

For this reason, heterotrophs are also known as consumers.

Consumers include all animals and fungi and many protists and bacteria.

They may consume autotrophs or other heterotrophs or organic molecules from other organisms.

Heterotrophs show great diversity but heterotrophs are limited by our utter dependence on those autotrophs that originally made our food.

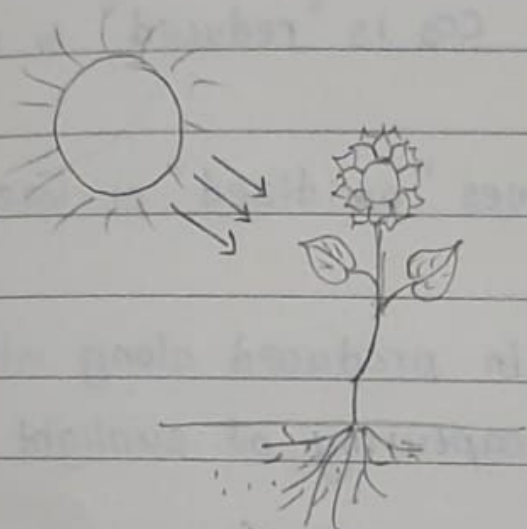
If plants, algae and autotrophic bacteria vanished from earth, animals, fungi, and other heterotrophs would soon disappear as well.

All life requires a constant input of energy.

Only autotrophs can transform that ultimate, solar source into the chemical energy in food that powers life.

Photosynthesis

Photosynthesis is the process used by plants, algae and certain bacteria to capture energy from sunlight and turn it into chemical energy.



The process of Photosynthesis.

There are two types of photosynthetic processes:

- i. Oxygenic photosynthesis.
- ii. anoxygenic photosynthesis.

The general principles are very similar, but oxygenic photosynthesis is the most common.

During Oxygenic photosynthesis.

light energy transfers electrons from water (H_2O) to carbon dioxide (CO_2), to produce carbohydrates.

In this transfer, the CO_2 is "reduced" or receives electrons,

and the water becomes "oxidized" or loses electrons.

Ultimately oxygen is produced along with carbohydrates. The capturing of sunlight is done by chloroplast

Anoxygenic photosynthesis does not produce oxygen hence the name is given as anoxygenic.

What is produced depends on the electron donor.

For example, many